CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD

CENTRAL VALLEY REGION
11020 SUN CENTER DR #200
RANCHO CORDOVA CA 95670-6114

2023 ANNUAL REPORT

PREPARED FOR:

JR BARROSO DAIRY

6902 E LE GRAND RD

MERCED CA 95341

NOTES:

Report submitted late due to financial hardships and sample results not being received from lab to be able to complete report.

PREPARED BY:

CAS
CARDOSO AG SERVICES

PO BOX 906
NEWMAN CA 95360

Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: Joe and Renee Barroso Dairy

Physical address of dairy:

Number and Street	City	County	Zip Code
6902 E Legrand RD	Merced	Merced	95341

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1960

Regional Water Quality Control Board Basin Plan designation: San Joaquin River Basin

County Assessor Parcel Number(s) for dairy facility:

X067-X110-X015-XXXX

B. OPERATORS

Barroso, David

Operator name: Barroso, David

Telephone no.: (209) 722-6437 Landline, (209) 652-6437 Cellular

Mailing Address Number and Street	City	State	Zip Code
6930 E Le Grand RD	Merced	CA	95341

This operator is responsible for paying permit fees.

C. OWNERS

Barroso, David

Legal owner name: Barroso, David

Telephone no.: (209) 722-6437 Landline, (209) 652-6437 Cellular

Mailing Address Number and Street	City	State	Zip Code
6930 E Le Grand RD	Merced	CA	95341

This owner is responsible for paying permit fees.

Barroso, Renee

Legal owner name:	Barroso, Renee	Telephone no.:	(209) 723-3191	(209) 652-6399
			Landline	Cellular
6902 E Le Grand RD	Merced		CA	95341
Mailing Address Number and Street	City		State	Zip Code

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	115	150	92	0	0
Number under roof	680	0	0	0	64	59
Maximum number	680	115	150	92	64	59
Average number	680	115	150	92	64	59
Avg live weight (lbs)	1,400	1,500	1,100	775		

Predominant milk cow breed: Holstein

Average milk production: 72 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 22,244.44 *tons per reporting period*

Total nitrogen from manure: 280,383.93 *lbs per reporting period*

After ammonia losses (30% loss applied): 196,268.75 *lbs per reporting period*

Total phosphorus from manure: 46,093.97 *lbs per reporting period*

Total potassium from manure: 131,119.79 *lbs per reporting period*

Total salt from manure: 346,622.25 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 55,440,000 *gallons*

Total nitrogen generated: 148,860.00 *lbs*

Total phosphorus generated: 23,935.90 *lbs*

Total potassium generated: 195,962.46 *lbs*

Total salt generated: 1,168,582.73 *lbs*

	55,440,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	55,440,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Canal	Surface water
Domestic Well	Ground water
Irrigation Well	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
1	21	21	2	manure	X067-X100-X054-XXXX
10	113	113	2	process wastewater	X670-X160-X019-XXXX
2	38	38	1	none	X067-X100-X054-XXXX
3	27	27	1	none	X067-X110-X015-XXXX
4	86	86	2	process wastewater	X067-X110-X015-XXXX
5	31	31	2	process wastewater	X067-X110-X015-XXXX
6	99	99	1	none	X067-X110-X015-XXXX
7	108	108	1	none	X067-X110-X015-XXXX
8	83	83	2	process wastewater	X067-X110-X015-XXXX
9	95	95	2	process wastewater	X067-X110-X016-XXXX
Totals for areas that were used for application	429	429	12		
Totals for areas that were not used for application	272	272	4		
Land application area totals	701	701	16		

B. CROPS AND HARVESTS

1

Field name: 1

10/30/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 21 Plant date: 10/30/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	235.00 *ton*	Dry-weight		65.4	22,600.00	3,100.00	22,400.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.19	175.01	24.01	173.46	844.08

1

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 21 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	567.00 *ton*	Dry-weight		65.7	12,200.00	2,300.00	13,700.00		6.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.00	225.97	42.60	253.75	1,222.45

10

Field name: 10

10/05/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 113 Plant date: 10/05/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,289.00 *ton*	Dry-weight		65.4	22,600.00	3,100.00	22,400.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.41	178.40	24.47	176.82	860.41

06/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 113 Plant date: 06/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	2,987.00 *ton*	Dry-weight		63.4	13,100.00	3,100.00	24,600.00		7.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.43	253.48	59.98	476.00	1,509.25

2

Field name: 2

12/01/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 38 Plant date: 12/01/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	296.00 *ton*	Dry-weight		72.3	29,500.00	3,100.00	13,800.00		8.30

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.79	127.30	13.38	59.55	358.18

3

Field name: 3

12/01/2019: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 27 Plant date: 12/01/2019

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/15/2023	208.00 *ton*	Dry-weight		8.3	27,300.00	3,000.00	38,400.00		11.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.70	385.71	42.39	542.54	1,638.92

4

Field name: 4

4

10/29/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 86 Plant date: 10/29/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,057.00 *ton*	Dry-weight		68.6	19,500.00	3,200.00	22,100.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	12.29	150.51	24.70	170.58	810.45

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 86 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/13/2023	2,357.00 *ton*	Dry-weight		59.7	12,400.00	2,100.00	17,500.00		7.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	27.41	273.92	46.39	386.58	1,656.75

5

Field name: 5

10/08/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 31 Plant date: 10/08/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	358.00 *ton*	Dry-weight		68.6	19,500.00	3,200.00	22,100.00		10.50

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.55	141.42	23.21	160.28	761.50

5

06/08/2023: Corn, silage

Crop: Corn, silage Acres planted: 31 Plant date: 06/08/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	750.00 *ton*	Dry-weight		70.0	14,000.00	3,000.00	9,700.00		5.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	24.19	203.23	43.55	140.81	856.45

6

Field name: 6

12/01/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 99 Plant date: 12/01/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/17/2023	779.00 *ton*	Dry-weight		4.1	37,800.00	2,800.00	14,600.00		8.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.87	570.48	42.26	220.35	1,343.20

7

Field name: 7

7

12/01/2018: Alfalfa, hay

Crop: Alfalfa, hay Acres planted: 108 Plant date: 12/01/2018

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/16/2023	825.00 *ton*	Dry-weight		8.0	35,600.00	3,300.00	24,300.00		10.20

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	8.00	480.00	43.20	336.00	0.00
Total actual harvest content	7.64	500.38	46.38	341.55	1,433.67

8

Field name: 8

10/10/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 83 Plant date: 10/10/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	984.00 *ton*	Dry-weight		74.3	22,000.00	3,100.00	21,700.00		12.80

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.86	134.06	18.89	132.23	779.99

06/10/2023: Corn, silage

Crop: Corn, silage Acres planted: 83 Plant date: 06/10/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	2,105.00 *ton*	Dry-weight		63.4	14,800.00	1,800.00	8,600.00		7.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	25.36	274.76	33.42	159.66	1,429.47

9

Field name: 9

10/01/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 95 Plant date: 10/01/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/10/2023	1,098.00 *ton*	Dry-weight		65.4	22,600.00	3,100.00	22,400.00		10.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	11.56	180.76	24.79	179.16	871.79

06/12/2023: Corn, silage

Crop: Corn, silage Acres planted: 95 Plant date: 06/12/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
10/04/2023	2,547.00 *ton*	Dry-weight		70.0	14,400.00	2,600.00	10,800.00		6.90

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	30.00	240.00	45.00	198.00	0.00
Total actual harvest content	26.81	231.64	41.82	173.73	1,109.96

NUTRIENT BUDGET

A. LAND APPLICATIONS

1 - 10/30/2022: Oats, silage-soft dough

Field name: 1

Crop: Oats, silage-soft dough Plant date: 10/30/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/28/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Separator solids	210.27	40.91	98.95	4,091.14	135.00 *ton*
Application event totals		210.27	40.91	98.95	4,091.14	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.84	0.00	0.00	0.00	2,055,944.00 *gal*
Application event totals		7.84	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.51	0.00	0.00	0.00	1,967,832.00 *gal*
Application event totals		7.51	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.62	0.00	0.00	0.00	1,997,203.00 *gal*
Application event totals		7.62	0.00	0.00	0.00	

1 - 06/08/2023: Corn, silage

Field name: 1

Crop: Corn, silage Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

1 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	15.24	0.00	0.00	0.00	3,994,406.00 *gal*
Application event totals		15.24	0.00	0.00	0.00	

05/19/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Solid Manure	Scraped material	199.08	71.60	220.03	0.00	480.00 *ton*
Application event totals		199.08	71.60	220.03	0.00	

06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	15.13	0.00	0.00	0.00	3,965,035.00 *gal*
Application event totals		15.13	0.00	0.00	0.00	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	15.24	0.00	0.00	0.00	3,994,406.00 *gal*
Application event totals		15.24	0.00	0.00	0.00	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	15.69	0.00	0.00	0.00	4,111,888.00 *gal*
Application event totals		15.69	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	16.02	0.00	0.00	0.00	4,200,000.00 *gal*
Application event totals		16.02	0.00	0.00	0.00	

1 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	15.57	0.00	0.00	0.00	4,082,517.00 *gal*
Application event totals		15.57	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	15.57	0.00	0.00	0.00	4,082,517.00 *gal*
Application event totals		15.57	0.00	0.00	0.00	

10 - 10/05/2022: Oats, silage-soft dough

Field name: 10

Crop: Oats, silage-soft dough

Plant date: 10/05/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	80.43	10.92	117.51	587.55	2,448,000.00 *gal*
Irrigation Well	Ground water	5.72	0.00	0.00	0.00	11,062,937.00 *gal*
Application event totals		86.15	10.92	117.51	587.55	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	83.19	13.97	97.84	617.64	2,304,000.00 *gal*
Irrigation Well	Ground water	5.47	0.00	0.00	0.00	10,588,811.00 *gal*
Application event totals		88.66	13.97	97.84	617.64	

10 - 10/05/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	5.56	0.00	0.00	0.00	10,746,853.00 *gal*
Application event totals		5.56	0.00	0.00	0.00	

10 - 06/15/2023: Corn, silage

Field name: 10

Crop: Corn, silage

Plant date: 06/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	86.75	13.07	104.81	585.10	2,016,000.00 *gal*
Irrigation Well	Ground water	11.11	0.00	0.00	0.00	21,493,706.00 *gal*
Application event totals		97.86	13.07	104.81	585.10	

06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.95	14.01	112.30	626.89	2,160,000.00 *gal*
Irrigation Well	Ground water	11.03	0.00	0.00	0.00	21,335,664.00 *gal*
Application event totals		103.98	14.01	112.30	626.89	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.11	0.00	0.00	0.00	21,493,706.00 *gal*
Application event totals		11.11	0.00	0.00	0.00	

10 - 06/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.03	4.65	33.68	256.82	2,520,000.00 *gal*
Irrigation Well	Ground water	11.44	0.00	0.00	0.00	22,125,874.00 *gal*
Application event totals		33.47	4.65	33.68	256.82	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.68	0.00	0.00	0.00	22,600,000.00 *gal*
Application event totals		11.68	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.96	6.11	44.27	337.53	3,312,000.00 *gal*
Irrigation Well	Ground water	11.36	0.00	0.00	0.00	21,967,832.00 *gal*
Application event totals		40.32	6.11	44.27	337.53	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	11.36	0.00	0.00	0.00	21,967,832.00 *gal*
Application event totals		11.36	0.00	0.00	0.00	

2 - 12/01/2019: Alfalfa, hay

Field name: 2

Crop: Alfalfa, hay

Plant date: 12/01/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following

2 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.86	0.00	0.00	23.97	3,898,716.00 *gal*
Application event totals		0.86	0.00	0.00	23.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	26.65	4,334,197.00 *gal*
Application event totals		0.95	0.00	0.00	26.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	27.56	4,482,686.00 *gal*
Application event totals		0.98	0.00	0.00	27.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.88	0.00	0.00	24.54	3,991,264.00 *gal*
Application event totals		0.88	0.00	0.00	24.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.92	0.00	0.00	25.65	4,171,199.00 *gal*
Application event totals		0.92	0.00	0.00	25.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.36	0.00	0.00	38.04	6,186,727.00 *gal*
Application event totals		1.36	0.00	0.00	38.04	

2 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.47	0.00	0.00	41.20	6,700,626.00 *gal*
Application event totals		1.47	0.00	0.00	41.20	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	26.49	4,307,395.00 *gal*
Application event totals		0.95	0.00	0.00	26.49	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	30.21	4,913,613.00 *gal*
Application event totals		1.08	0.00	0.00	30.21	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	35.38	5,753,918.00 *gal*
Application event totals		1.26	0.00	0.00	35.38	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	33.67	5,475,911.00 *gal*
Application event totals		1.20	0.00	0.00	33.67	

10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.17	0.00	0.00	32.85	5,342,963.00 *gal*
Application event totals		1.17	0.00	0.00	32.85	

3 - 12/01/2019: Alfalfa, hay

3 - 12/01/2019: Alfalfa, hay

Field name: 3

Crop: Alfalfa, hay

Plant date: 12/01/2019

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.86	0.00	0.00	23.97	2,770,140.00 *gal*
Application event totals		0.86	0.00	0.00	23.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	26.65	3,079,561.00 *gal*
Application event totals		0.95	0.00	0.00	26.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	27.56	3,185,066.00 *gal*
Application event totals		0.98	0.00	0.00	27.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.88	0.00	0.00	24.54	2,835,898.00 *gal*
Application event totals		0.88	0.00	0.00	24.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.92	0.00	0.00	25.65	2,963,747.00 *gal*
Application event totals		0.92	0.00	0.00	25.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.36	0.00	0.00	38.04	4,395,833.00 *gal*
Application event totals		1.36	0.00	0.00	38.04	

3 - 12/01/2019: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.47	0.00	0.00	41.20	4,760,971.00 *gal*
Application event totals		1.47	0.00	0.00	41.20	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	26.49	3,060,517.00 *gal*
Application event totals		0.95	0.00	0.00	26.49	

08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	30.21	3,491,252.00 *gal*
Application event totals		1.08	0.00	0.00	30.21	

08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	35.38	4,088,310.00 *gal*
Application event totals		1.26	0.00	0.00	35.38	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	33.67	3,890,779.00 *gal*
Application event totals		1.20	0.00	0.00	33.67	

10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.17	0.00	0.00	32.85	3,796,316.00 *gal*
Application event totals		1.17	0.00	0.00	32.85	

4 - 10/29/2022: Oats, silage-soft dough

4 - 10/29/2022: Oats, silage-soft dough

Field name: 4

Crop: Oats, silage-soft dough

Plant date: 10/29/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
11/19/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.71	10.55	113.53	567.65	1,800,000.00 *gal*
Irrigation Well	Ground water	8.09	0.00	0.00	0.00	8,419,580.00 *gal*
Application event totals		85.80	10.55	113.53	567.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.15	12.62	88.38	557.94	1,584,000.00 *gal*
Irrigation Well	Ground water	7.74	0.00	0.00	0.00	8,058,741.00 *gal*
Application event totals		82.89	12.62	88.38	557.94	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/01/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	7.86	0.00	0.00	0.00	8,179,021.00 *gal*
Application event totals		7.86	0.00	0.00	0.00	

4 - 06/08/2023: Corn, silage

Field name: 4

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/20/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	15.71	0.00	0.00	0.00	16,358,042.00 *gal*
Application event totals		15.71	0.00	0.00	0.00	

4 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.70	14.72	118.04	658.97	1,728,000.00 *gal*
Irrigation Well	Ground water	15.60	0.00	0.00	0.00	16,237,762.00 *gal*
Application event totals		113.30	14.72	118.04	658.97	

07/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.95	6.11	44.26	337.45	2,520,000.00 *gal*
Irrigation Well	Ground water	15.71	0.00	0.00	0.00	16,358,042.00 *gal*
Application event totals		44.67	6.11	44.26	337.45	

07/23/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	16.18	0.00	0.00	0.00	16,839,161.00 *gal*
Application event totals		16.18	0.00	0.00	0.00	

08/05/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	24.82	5.24	37.94	289.24	2,160,000.00 *gal*
Irrigation Well	Ground water	16.52	0.00	0.00	0.00	17,200,000.00 *gal*
Application event totals		41.34	5.24	37.94	289.24	

08/19/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.78	6.29	45.52	347.09	2,592,000.00 *gal*
Irrigation Well	Ground water	16.06	0.00	0.00	0.00	16,718,881.00 *gal*
Application event totals		45.84	6.29	45.52	347.09	

4 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/02/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	16.06	0.00	0.00	0.00	16,718,881.00 *gal*
Application event totals		16.06	0.00	0.00	0.00	

5 - 10/08/2022: Oats, silage-soft dough

Field name: 5

Crop: Oats, silage-soft dough

Plant date: 10/08/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	77.61	10.54	113.38	566.92	648,000.00 *gal*
Irrigation Well	Ground water	4.66	0.00	0.00	0.00	3,034,965.00 *gal*
Application event totals		82.26	10.54	113.38	566.92	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	75.81	12.73	89.16	562.85	576,000.00 *gal*
Irrigation Well	Ground water	4.46	0.00	0.00	0.00	2,904,895.00 *gal*
Application event totals		80.26	12.73	89.16	562.85	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.52	0.00	0.00	0.00	2,948,252.00 *gal*
Application event totals		4.52	0.00	0.00	0.00	

5 - 06/08/2023: Corn, silage

5 - 06/08/2023: Corn, silage

Field name: 5

Crop: Corn, silage

Plant date: 06/08/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.05	0.00	0.00	0.00	5,896,503.00 *gal*
Application event totals		9.05	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	96.00	14.46	115.98	647.45	612,000.00 *gal*
Irrigation Well	Ground water	8.98	0.00	0.00	0.00	5,853,147.00 *gal*
Application event totals		104.98	14.46	115.98	647.45	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	28.69	6.06	43.85	334.34	900,000.00 *gal*
Irrigation Well	Ground water	9.05	0.00	0.00	0.00	5,896,503.00 *gal*
Application event totals		37.73	6.06	43.85	334.34	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.31	0.00	0.00	0.00	6,069,930.00 *gal*
Application event totals		9.31	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.24	5.33	38.59	294.22	792,000.00 *gal*
Irrigation Well	Ground water	9.51	0.00	0.00	0.00	6,200,000.00 *gal*
Application event totals		34.76	5.33	38.59	294.22	

5 - 06/08/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	29.83	6.30	45.61	347.71	936,000.00 *gal*
Irrigation Well	Ground water	9.25	0.00	0.00	0.00	6,026,573.00 *gal*
Application event totals		39.08	6.30	45.61	347.71	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.25	0.00	0.00	0.00	6,026,573.00 *gal*
Application event totals		9.25	0.00	0.00	0.00	

6 - 12/01/2018: Alfalfa, hay

Field name: 6

Crop: Alfalfa, hay

Plant date: 12/01/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.86	0.00	0.00	23.97	10,157,182.00 *gal*
Application event totals		0.86	0.00	0.00	23.97	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	26.65	11,291,723.00 *gal*
Application event totals		0.95	0.00	0.00	26.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	27.56	11,678,576.00 *gal*
Application event totals		0.98	0.00	0.00	27.56	

6 - 12/01/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.88	0.00	0.00	24.54	10,398,292.00 *gal*
Application event totals		0.88	0.00	0.00	24.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.92	0.00	0.00	25.65	10,867,071.00 *gal*
Application event totals		0.92	0.00	0.00	25.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.36	0.00	0.00	38.04	16,118,053.00 *gal*
Application event totals		1.36	0.00	0.00	38.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.47	0.00	0.00	41.20	17,456,895.00 *gal*
Application event totals		1.47	0.00	0.00	41.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	26.49	11,221,897.00 *gal*
Application event totals		0.95	0.00	0.00	26.49	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	30.21	12,801,256.00 *gal*
Application event totals		1.08	0.00	0.00	30.21	

6 - 12/01/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	35.38	14,990,470.00 *gal*
Application event totals		1.26	0.00	0.00	35.38	

09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	33.67	14,266,189.00 *gal*
Application event totals		1.20	0.00	0.00	33.67	

10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.17	0.00	0.00	32.85	13,919,824.00 *gal*
Application event totals		1.17	0.00	0.00	32.85	

7 - 12/01/2018: Alfalfa, hay

Field name: 7

Crop: Alfalfa, hay

Plant date: 12/01/2018

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
04/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.86	0.00	0.00	23.97	11,080,562.00 *gal*
Application event totals		0.86	0.00	0.00	23.97	

04/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	26.65	12,318,244.00 *gal*
Application event totals		0.95	0.00	0.00	26.65	

7 - 12/01/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.98	0.00	0.00	27.56	12,740,265.00 *gal*
Application event totals		0.98	0.00	0.00	27.56	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/09/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.88	0.00	0.00	24.54	11,343,592.00 *gal*
Application event totals		0.88	0.00	0.00	24.54	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.92	0.00	0.00	25.65	11,854,986.00 *gal*
Application event totals		0.92	0.00	0.00	25.65	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.36	0.00	0.00	38.04	17,583,331.00 *gal*
Application event totals		1.36	0.00	0.00	38.04	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/11/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.47	0.00	0.00	41.20	19,043,885.00 *gal*
Application event totals		1.47	0.00	0.00	41.20	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	0.95	0.00	0.00	26.49	12,242,069.00 *gal*
Application event totals		0.95	0.00	0.00	26.49	

7 - 12/01/2018: Alfalfa, hay

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/08/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.08	0.00	0.00	30.21	13,965,007.00 *gal*
Application event totals		1.08	0.00	0.00	30.21	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/21/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.26	0.00	0.00	35.38	16,353,240.00 *gal*
Application event totals		1.26	0.00	0.00	35.38	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.20	0.00	0.00	33.67	15,563,115.00 *gal*
Application event totals		1.20	0.00	0.00	33.67	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Canal	Surface water	1.17	0.00	0.00	32.85	15,185,263.00 *gal*
Application event totals		1.17	0.00	0.00	32.85	

8 - 10/10/2022: Oats, silage-soft dough

Field name: 8

Crop: Oats, silage-soft dough

Plant date: 10/10/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	74.07	10.06	108.22	541.12	1,656,000.00 *gal*
Irrigation Well	Ground water	5.15	0.00	0.00	0.00	8,125,874.00 *gal*
Application event totals		79.22	10.06	108.22	541.12	

8 - 10/10/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.93	0.00	0.00	0.00	7,777,622.00 *gal*
Application event totals		4.93	0.00	0.00	0.00	

04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.36	12.71	101.93	568.99	1,440,000.00 *gal*
Irrigation Well	Ground water	5.00	0.00	0.00	0.00	7,893,706.00 *gal*
Application event totals		89.36	12.71	101.93	568.99	

8 - 06/10/2023: Corn, silage

Field name: 8

Crop: Corn, silage

Plant date: 06/10/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/25/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	97.02	14.62	117.21	654.34	1,656,000.00 *gal*
Irrigation Well	Ground water	10.00	0.00	0.00	0.00	15,787,413.00 *gal*
Application event totals		107.02	14.62	117.21	654.34	

06/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	9.93	0.00	0.00	0.00	15,671,329.00 *gal*
Application event totals		9.93	0.00	0.00	0.00	

8 - 06/10/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.00	6.33	45.86	349.65	2,520,000.00 *gal*
Irrigation Well	Ground water	10.00	0.00	0.00	0.00	15,787,413.00 *gal*
Application event totals		40.00	6.33	45.86	349.65	

07/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.29	0.00	0.00	0.00	16,251,748.00 *gal*
Application event totals		10.29	0.00	0.00	0.00	

08/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	25.71	5.43	39.31	299.70	2,160,000.00 *gal*
Irrigation Well	Ground water	10.51	0.00	0.00	0.00	16,600,000.00 *gal*
Application event totals		36.23	5.43	39.31	299.70	

08/27/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	30.86	6.52	47.17	359.64	2,592,000.00 *gal*
Irrigation Well	Ground water	10.22	0.00	0.00	0.00	16,135,664.00 *gal*
Application event totals		41.08	6.52	47.17	359.64	

09/17/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	10.22	0.00	0.00	0.00	16,135,664.00 *gal*
Application event totals		10.22	0.00	0.00	0.00	

9 - 10/01/2022: Oats, silage-soft dough

Field name: 9

Crop: Oats, silage-soft dough

Plant date: 10/01/2022

9 - 10/01/2022: Oats, silage-soft dough

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/23/2022	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	73.16	9.93	106.89	534.43	1,872,000.00 *gal*
Irrigation Well	Ground water	4.33	0.00	0.00	0.00	9,300,699.00 *gal*
Application event totals		77.49	9.93	106.89	534.43	

03/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	4.14	0.00	0.00	0.00	8,902,098.00 *gal*
Application event totals		4.14	0.00	0.00	0.00	

04/04/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	88.45	13.33	106.86	596.54	1,728,000.00 *gal*
Irrigation Well	Ground water	4.21	0.00	0.00	0.00	9,034,965.00 *gal*
Application event totals		92.66	13.33	106.86	596.54	

9 - 06/12/2023: Corn, silage

Field name: 9

Crop: Corn, silage

Plant date: 06/12/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/24/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	84.76	12.77	102.41	571.68	1,656,000.00 *gal*
Irrigation Well	Ground water	8.41	0.00	0.00	0.00	18,069,930.00 *gal*
Application event totals		93.18	12.77	102.41	571.68	

9 - 06/12/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	92.13	13.88	111.31	621.40	1,800,000.00 *gal*
Irrigation Well	Ground water	8.35	0.00	0.00	0.00	17,937,063.00 *gal*
Application event totals		100.48	13.88	111.31	621.40	

07/12/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.41	0.00	0.00	0.00	18,069,930.00 *gal*
Application event totals		8.41	0.00	0.00	0.00	

07/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	22.47	4.74	34.34	261.84	2,160,000.00 *gal*
Irrigation Well	Ground water	8.66	0.00	0.00	0.00	18,601,399.00 *gal*
Application event totals		31.13	4.74	34.34	261.84	

08/13/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Water	Process wastewater	26.96	5.69	41.21	314.21	2,592,000.00 *gal*
Irrigation Well	Ground water	8.85	0.00	0.00	0.00	19,000,000.00 *gal*
Application event totals		35.80	5.69	41.21	314.21	

08/28/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.60	0.00	0.00	0.00	18,468,531.00 *gal*
Application event totals		8.60	0.00	0.00	0.00	

9 - 06/12/2023: Corn, silage

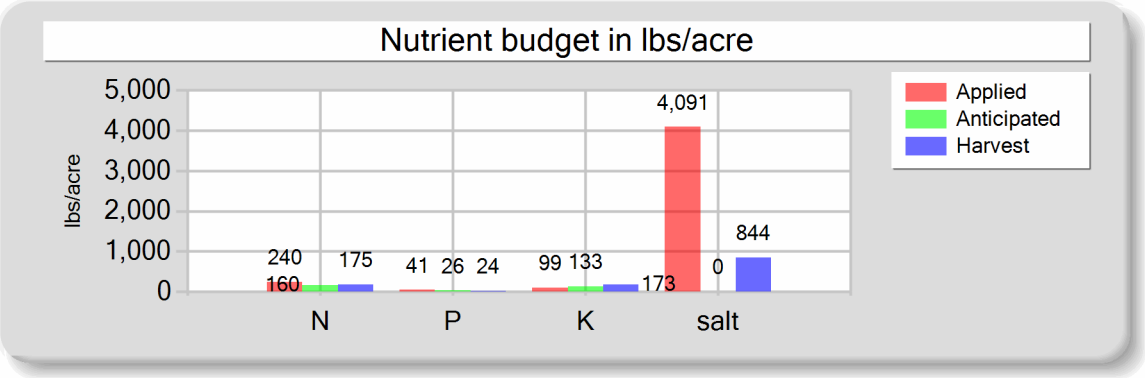
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
09/18/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Irrigation Well	Ground water	8.60	0.00	0.00	0.00	18,468,531.00 *gal*
Application event totals		8.60	0.00	0.00	0.00	

B. NUTRIENT BUDGET

1 - 10/30/2022: Oats, silage-soft dough

Field name: 1 Crop: Oats, silage-soft dough Plant date: 10/30/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	210.27	40.91	98.95	4,091.14
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	22.97	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	240.23	40.91	98.95	4,091.14
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	175.01	24.01	173.46	844.08
Nutrient balance	65.22	16.91	-74.51	3,247.07
Applied to removed ratio	1.37	1.70	0.57	4.85

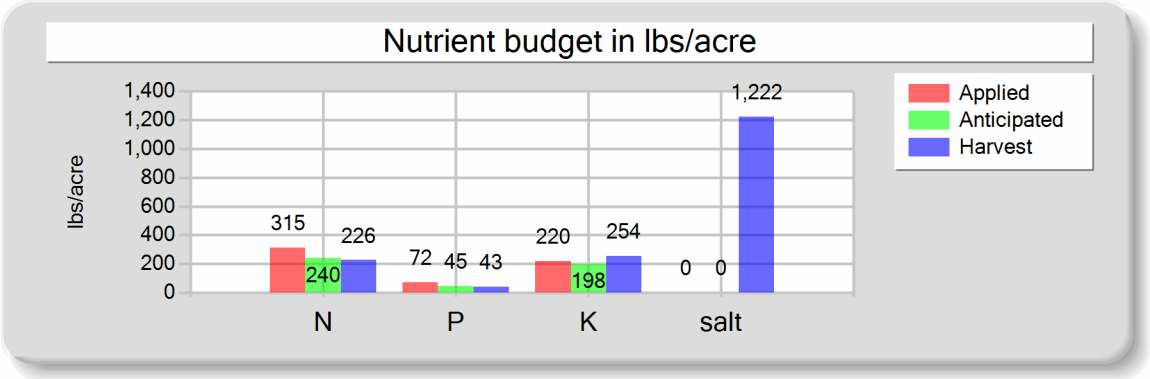
Fresh water applied	
6,020,979.00	*gallons*
221.73	*acre-inches*
10.56	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

1 - 06/08/2023: Corn, silage

Field name: 1 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	199.08	71.60	220.03	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	108.46	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	314.54	71.60	220.03	0.00
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	225.97	42.60	253.75	1,222.45
Nutrient balance	88.57	29.00	-33.72	-1,222.45
Applied to removed ratio	1.39	1.68	0.87	0.00

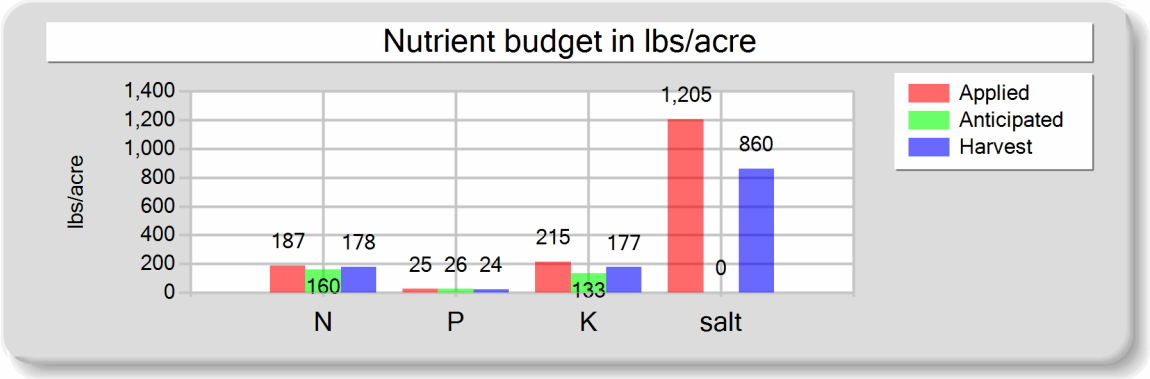
Fresh water applied	
28,430,769.00	*gallons*
1,047.01	*acre-inches*
49.86	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 10/05/2022: Oats, silage-soft dough

Field name: 10 Crop: Oats, silage-soft dough Plant date: 10/05/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	163.62	24.89	215.35	1,205.19
Fresh water	16.75	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	187.37	24.89	215.35	1,205.19
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	178.40	24.47	176.82	860.41
Nutrient balance	8.97	0.42	38.53	344.78
Applied to removed ratio	1.05	1.02	1.22	1.40

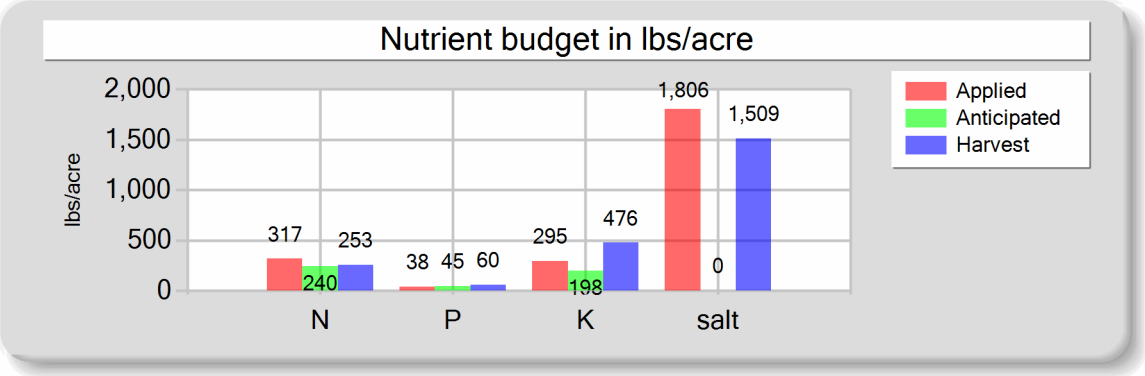
Fresh water applied	
32,398,601.00	*gallons*
1,193.13	*acre-inches*
10.56	*inches/acre*

Process wastewater applied	
4,752,000.00	*gallons*
175.00	*acre-inches*
1.55	*inches/acre*

Total harvests for the crop	
1	*harvests*

10 - 06/15/2023: Corn, silage

Field name: 10 Crop: Corn, silage Plant date: 06/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	230.70	37.84	295.07	1,806.35
Fresh water	79.08	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	316.78	37.84	295.07	1,806.35
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	253.48	59.98	476.00	1,509.25
Nutrient balance	63.30	-22.14	-180.93	297.09
Applied to removed ratio	1.25	0.63	0.62	1.20

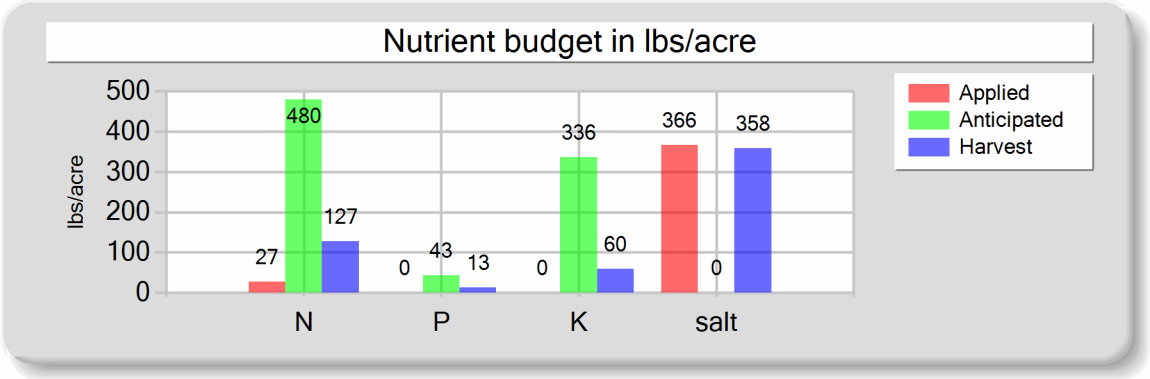
Fresh water applied	
152,984,614.00	*gallons*
5,633.90	*acre-inches*
49.86	*inches/acre*

Process wastewater applied	
10,008,000.00	*gallons*
368.56	*acre-inches*
3.26	*inches/acre*

Total harvests for the crop	
1	*harvests*

2 - 12/01/2019: Alfalfa, hay

Field name: 2 Crop: Alfalfa, hay Plant date: 12/01/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.08	0.00	0.00	366.23
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	27.08	0.00	0.00	366.23
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	127.30	13.38	59.55	358.18
Nutrient balance	-100.22	-13.38	-59.55	8.05
Applied to removed ratio	0.21	0.00	0.00	1.02

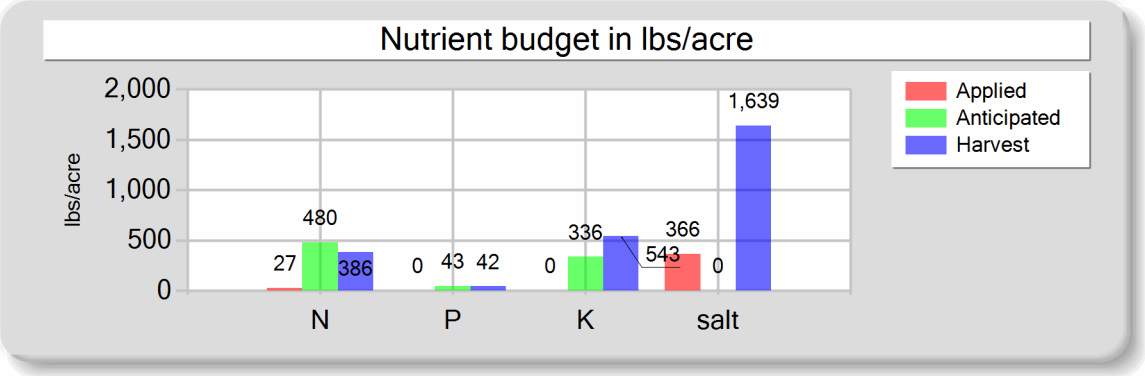
Fresh water applied	
59,559,215.00	*gallons*
2,193.36	*acre-inches*
57.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

3 - 12/01/2019: Alfalfa, hay

Field name: 3 Crop: Alfalfa, hay Plant date: 12/01/2019



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.08	0.00	0.00	366.23
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	27.08	0.00	0.00	366.23
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	385.71	42.39	542.54	1,638.92
Nutrient balance	-358.63	-42.39	-542.54	-1,272.69
Applied to removed ratio	0.07	0.00	0.00	0.22

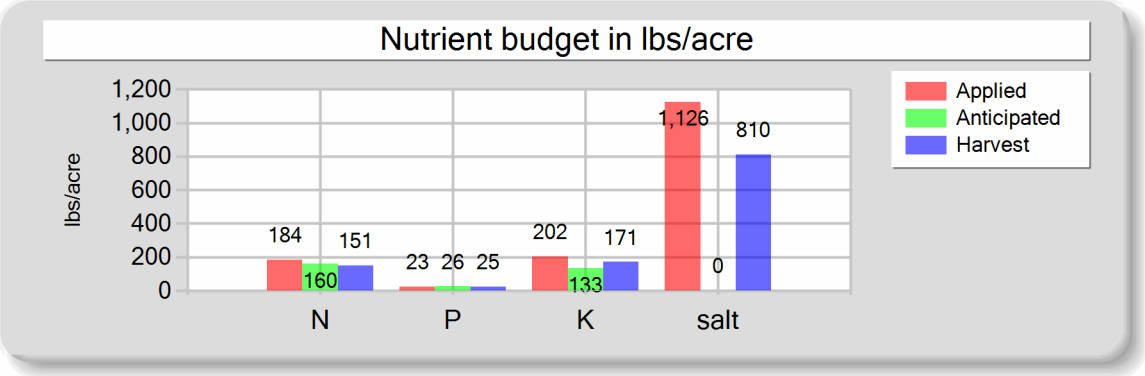
Fresh water applied	
42,318,390.00	*gallons*
1,558.44	*acre-inches*
57.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 10/29/2022: Oats, silage-soft dough

Field name: 4 Crop: Oats, silage-soft dough Plant date: 10/29/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	152.85	23.17	201.91	1,125.60
Fresh water	23.69	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	183.54	23.17	201.91	1,125.60
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	150.51	24.70	170.58	810.45
Nutrient balance	33.03	-1.53	31.33	315.15
Applied to removed ratio	1.22	0.94	1.18	1.39

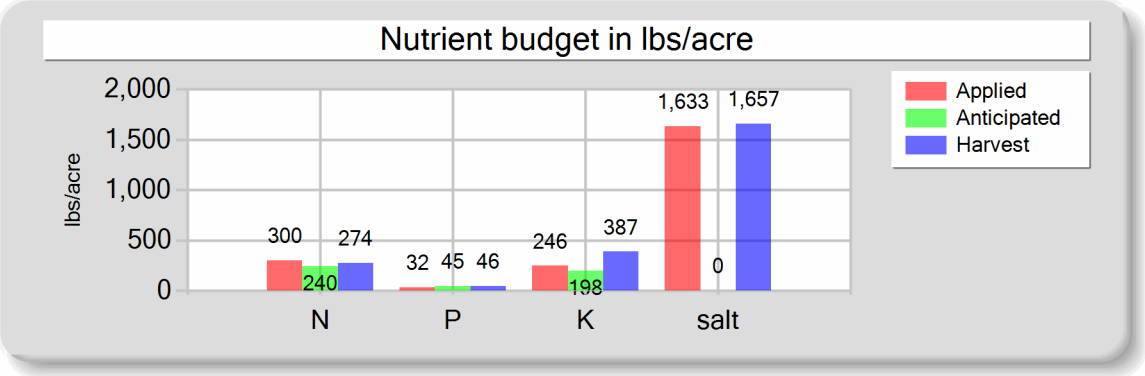
Fresh water applied	
24,657,342.00	*gallons*
908.05	*acre-inches*
10.56	*inches/acre*

Process wastewater applied	
3,384,000.00	*gallons*
124.62	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

4 - 06/08/2023: Corn, silage

Field name: 4 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	181.25	32.36	245.76	1,632.75
Fresh water	111.85	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	300.10	32.36	245.76	1,632.75
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	273.92	46.39	386.58	1,656.75
Nutrient balance	26.18	-14.03	-140.81	-24.00
Applied to removed ratio	1.10	0.70	0.64	0.99

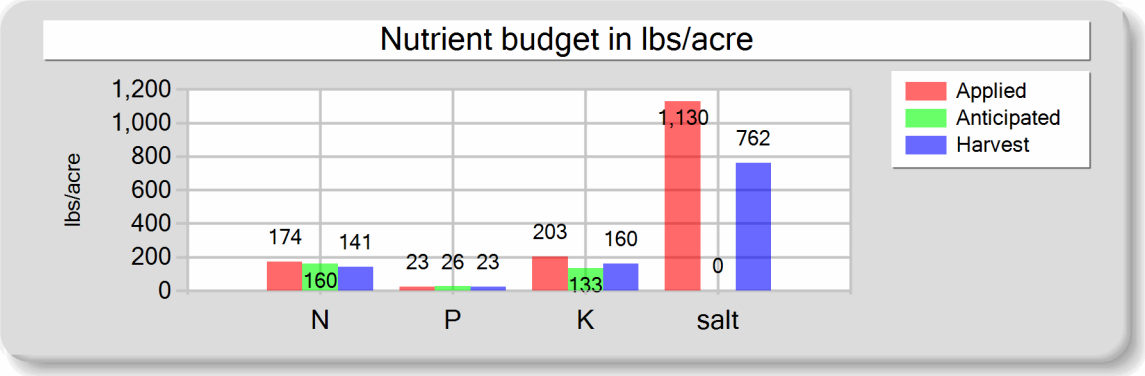
Fresh water applied	
116,430,769.00	*gallons*
4,287.75	*acre-inches*
49.86	*inches/acre*

Process wastewater applied	
9,000,000.00	*gallons*
331.44	*acre-inches*
3.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 10/08/2022: Oats, silage-soft dough

Field name: 5 Crop: Oats, silage-soft dough Plant date: 10/08/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	153.41	23.27	202.54	1,129.77
Fresh water	13.64	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	174.05	23.27	202.54	1,129.77
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	141.42	23.21	160.28	761.50
Nutrient balance	32.63	0.06	42.26	368.27
Applied to removed ratio	1.23	1.00	1.26	1.48

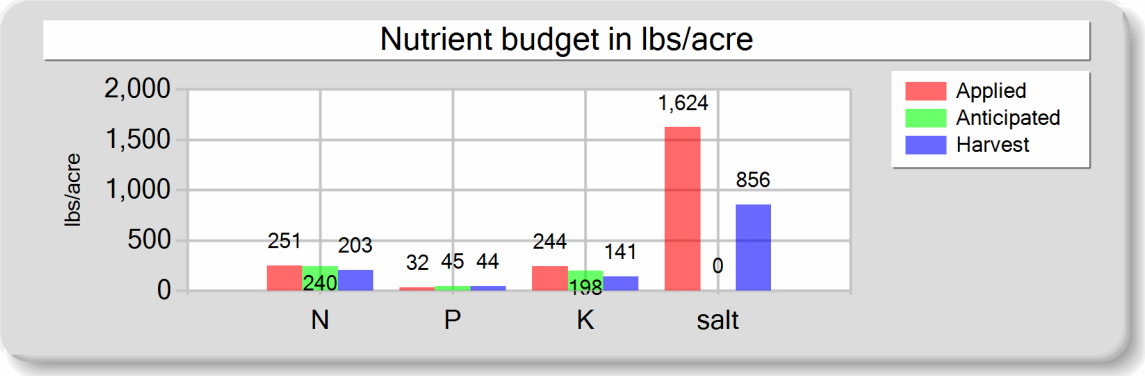
Fresh water applied	
8,888,112.00	*gallons*
327.32	*acre-inches*
10.56	*inches/acre*

Process wastewater applied	
1,224,000.00	*gallons*
45.08	*acre-inches*
1.45	*inches/acre*

Total harvests for the crop	
1	*harvests*

5 - 06/08/2023: Corn, silage

Field name: 5 Crop: Corn, silage Plant date: 06/08/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	179.76	32.15	244.03	1,623.72
Fresh water	64.40	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	251.16	32.15	244.03	1,623.72
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	203.23	43.55	140.81	856.45
Nutrient balance	47.93	-11.40	103.22	767.27
Applied to removed ratio	1.24	0.74	1.73	1.90

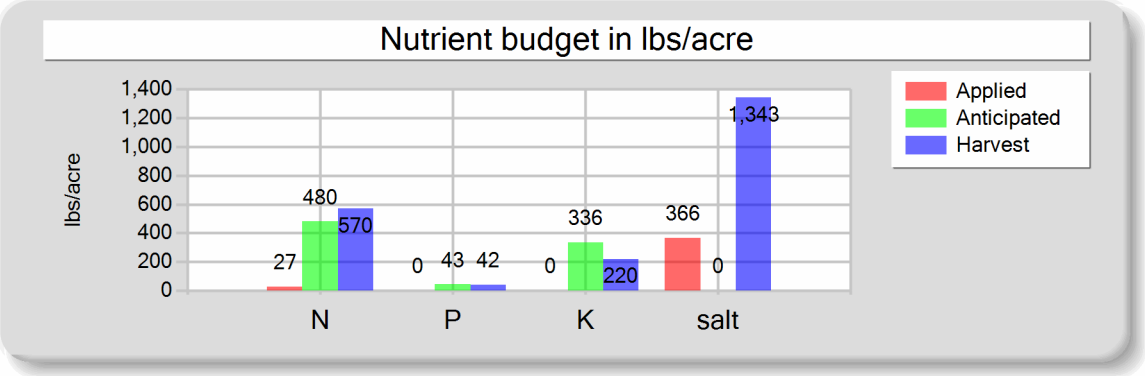
Fresh water applied	
41,969,229.00	*gallons*
1,545.58	*acre-inches*
49.86	*inches/acre*

Process wastewater applied	
3,240,000.00	*gallons*
119.32	*acre-inches*
3.85	*inches/acre*

Total harvests for the crop	
1	*harvests*

6 - 12/01/2018: Alfalfa, hay

Field name: 6 Crop: Alfalfa, hay Plant date: 12/01/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.08	0.00	0.00	366.23
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	27.08	0.00	0.00	366.23
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	570.48	42.26	220.35	1,343.20
Nutrient balance	-543.40	-42.26	-220.35	-976.97
Applied to removed ratio	0.05	0.00	0.00	0.27

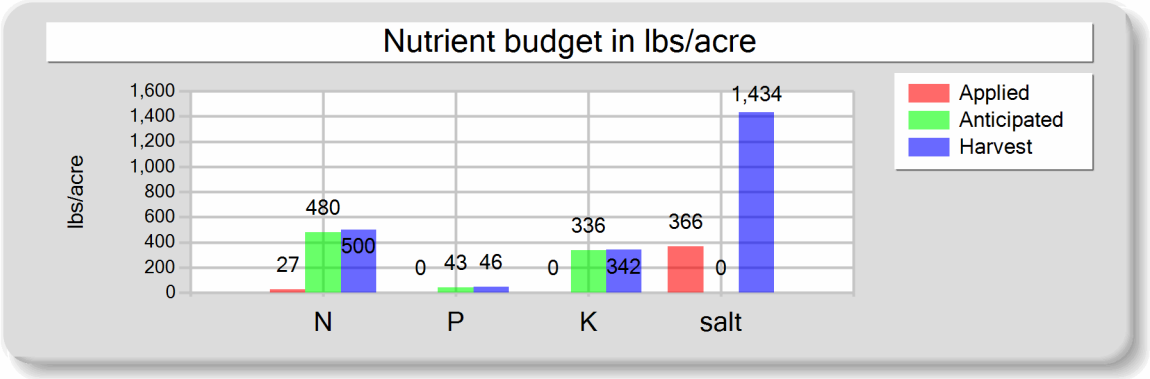
Fresh water applied	
155,167,428.00	*gallons*
5,714.29	*acre-inches*
57.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

7 - 12/01/2018: Alfalfa, hay

Field name: 7 Crop: Alfalfa, hay Plant date: 12/01/2018



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	13.08	0.00	0.00	366.23
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	27.08	0.00	0.00	366.23
Anticipated crop nutrient removal	480.00	43.20	336.00	0.00
Actual crop nutrient removal	500.38	46.38	341.55	1,433.67
Nutrient balance	-473.30	-46.38	-341.55	-1,067.44
Applied to removed ratio	0.05	0.00	0.00	0.26

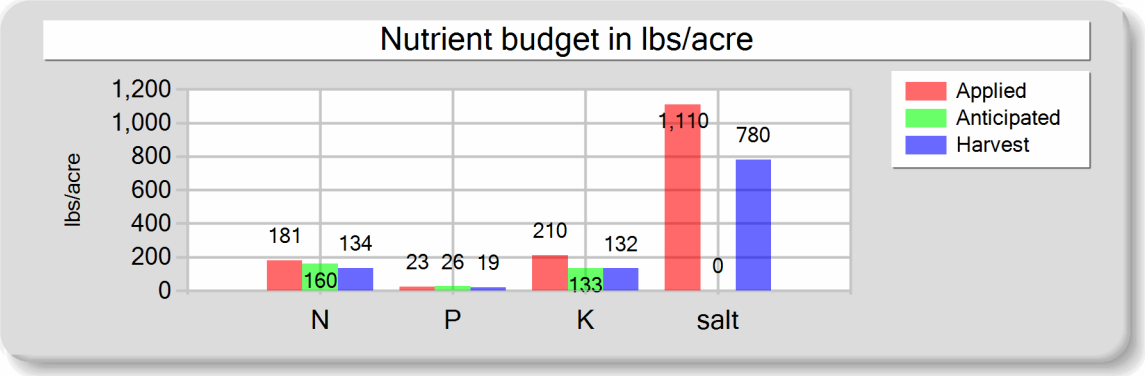
Fresh water applied	
169,273,559.00	*gallons*
6,233.77	*acre-inches*
57.72	*inches/acre*

Process wastewater applied	
0.00	*gallons*
0.00	*acre-inches*
0.00	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 10/10/2022: Oats, silage-soft dough

Field name: 8 Crop: Oats, silage-soft dough Plant date: 10/10/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	158.44	22.77	210.15	1,110.11
Fresh water	15.07	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	180.51	22.77	210.15	1,110.11
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	134.06	18.89	132.23	779.99
Nutrient balance	46.45	3.88	77.92	330.11
Applied to removed ratio	1.35	1.21	1.59	1.42

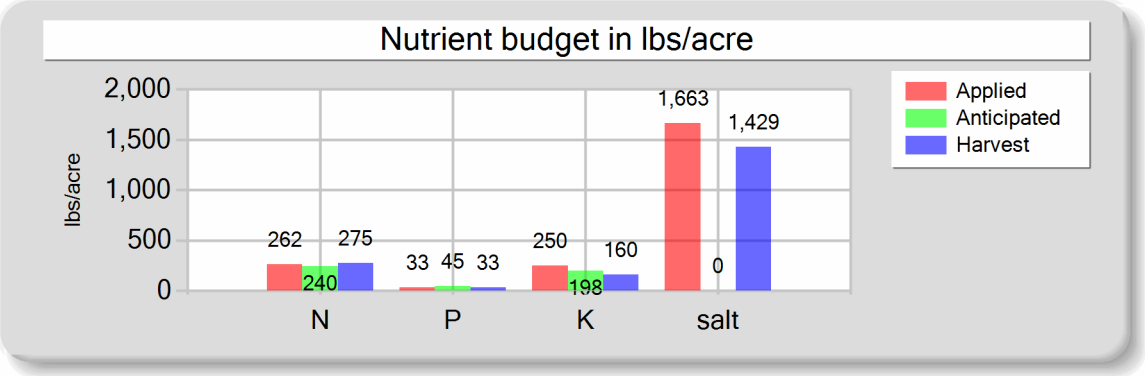
Fresh water applied	
23,797,202.00	*gallons*
876.37	*acre-inches*
10.56	*inches/acre*

Process wastewater applied	
3,096,000.00	*gallons*
114.02	*acre-inches*
1.37	*inches/acre*

Total harvests for the crop	
1	*harvests*

8 - 06/10/2023: Corn, silage

Field name: 8 Crop: Corn, silage Plant date: 06/10/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	183.59	32.90	249.55	1,663.31
Fresh water	71.18	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	261.76	32.90	249.55	1,663.31
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	274.76	33.42	159.66	1,429.47
Nutrient balance	-12.99	-0.52	89.90	233.84
Applied to removed ratio	0.95	0.98	1.56	1.16

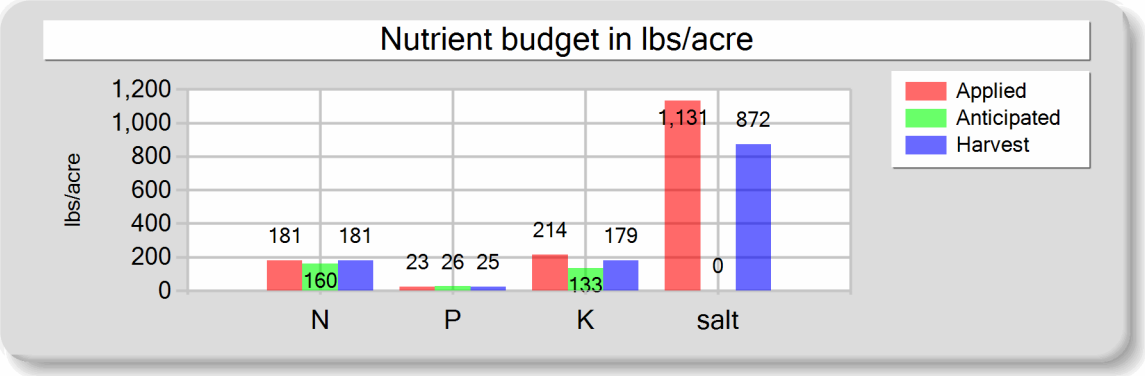
Fresh water applied	
112,369,231.00	*gallons*
4,138.18	*acre-inches*
49.86	*inches/acre*

Process wastewater applied	
8,928,000.00	*gallons*
328.79	*acre-inches*
3.96	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 10/01/2022: Oats, silage-soft dough

Field name: 9 Crop: Oats, silage-soft dough Plant date: 10/01/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	161.61	23.26	213.75	1,130.97
Fresh water	12.68	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	181.29	23.26	213.75	1,130.97
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	180.76	24.79	179.16	871.79
Nutrient balance	0.53	-1.53	34.59	259.18
Applied to removed ratio	1.00	0.94	1.19	1.30

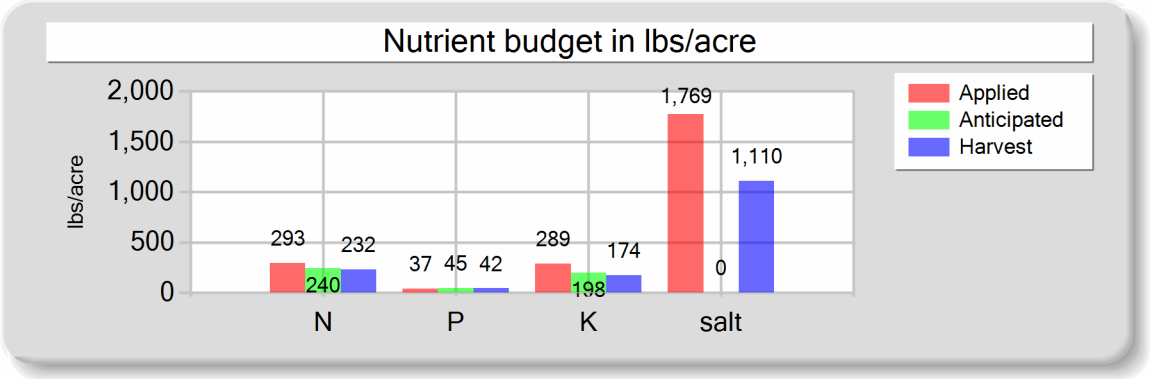
Fresh water applied	
27,237,762.00	*gallons*
1,003.07	*acre-inches*
10.56	*inches/acre*

Process wastewater applied	
3,600,000.00	*gallons*
132.58	*acre-inches*
1.40	*inches/acre*

Total harvests for the crop	
1	*harvests*

9 - 06/12/2023: Corn, silage

Field name: 9 Crop: Corn, silage Plant date: 06/12/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	0.00	0.00	0.00	0.00
Process wastewater	226.32	37.09	289.28	1,769.13
Fresh water	59.88	0.00	0.00	0.00
Atmospheric deposition	7.00	0.00	0.00	0.00
Total nutrients applied	293.20	37.09	289.28	1,769.13
Anticipated crop nutrient removal	240.00	45.00	198.00	0.00
Actual crop nutrient removal	231.64	41.82	173.73	1,109.96
Nutrient balance	61.56	-4.73	115.54	659.17
Applied to removed ratio	1.27	0.89	1.67	1.59

Fresh water applied	
128,615,384.00	*gallons*
4,736.47	*acre-inches*
49.86	*inches/acre*

Process wastewater applied	
8,208,000.00	*gallons*
302.27	*acre-inches*
3.18	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

SSB Manure - 2022

Sample and source description: SSB Manure - 2022

Sample date: 10/26/2022 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 26.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,100.00	4,300.00	10,400.00							43.00
DL	100.00	100.00	100.00							1.00

SSB Manure

Sample and source description: SSB Manure

Sample date: 05/10/2023 Material type: Separator solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	23,100.00	5,900.00	7,900.00							0.00
DL	0.01	0.01	0.01							1.00

Scraped Manure

Sample and source description: Scraped Manure

Sample date: 12/13/2023 Material type: Scraped material Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 61.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,400.00	4,100.00	12,600.00							0.00
DL	0.01	0.01	0.01							1.00

B. PROCESS WASTEWATER ANALYSES

2022 LAGOON Q4

Sample and source description: 2022 LAGOON Q4

Sample date: 12/01/2022 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.80

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	443.00	400.00	0.00	1.90	60.40	650.00								7,020.00	3,250
DL	0.70	0.20	0.01	0.02	0.02	0.20								2.00	10

Lagoon Q1

Sample and source description: Lagoon Q1

Sample date: 03/30/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	488.00	355.00	0.00	0.90	82.10	575.00								6,990.00	3,630
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q2

Sample and source description: Lagoon Q2

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	581.00	425.00	0.00	1.70	87.80	704.00								7,960.00	3,930
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q3

Sample and source description: Lagoon Q3

Sample date: 08/01/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.30

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	118.00	93.70	0.00	0.40	25.00	181.00								2,190.00	1,380
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

Lagoon Q4

Sample and source description: Lagoon Q4

Sample date: 12/13/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.40

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	444.00	316.00	0.00	1.30	87.00	732.00								6,030.00	2,850
DL	0.01	0.01	0.01	0.01	0.01	0.01								10.00	1

C. FRESH WATER ANALYSES

Canal

Canal

Sample description: Canal

Sample date: 10/19/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.00	0.00	1.00								32.20	28
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW House

Sample description: DW House

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	13.10	0.00	13.10								703.00	0
DL	0.01	0.01	0.01								10.00	1

Domestic Well

DW MB

Sample description: DW MB

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.10	0.00	5.10								474.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW F10

Sample description: IW F10

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	7.00	0.00	7.00								923.00	0
DL	0.01	0.01	0.01								10.00	1

IW F4

Sample description: IW F4

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.90	0.00	9.90								1,520.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW F6

Sample description: IW F6

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	9.60	0.00	9.60								1,240.00	0
DL	0.01	0.01	0.01								10.00	1

IW F7

Sample description: IW F7

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.50	0.00	5.50								553.00	0
DL	0.01	0.01	0.01								10.00	1

IW F8 North

Sample description: IW F8 North

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.70	0.00	5.70								581.00	0
DL	0.01	0.01	0.01								10.00	1

Irrigation Well

IW F8 South

Sample description: IW F8 South

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	6.30	0.00	6.30								1,420.00	0
DL	0.01	0.01	0.01								10.00	1

IW F9

Sample description: IW F9

Sample date: 12/13/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	5.30	0.00	5.30								522.00	0
DL	0.01	0.01	0.01								10.00	1

D. SOIL ANALYSES

No soil analyses entered.

E. PLANT TISSUE ANALYSES

1 - 10/30/2022: Oats, silage-soft dough

Field 1 Oats

Sample and source description: Field 1 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	3,100.00	22,400.00		10.90
DL	0.01	0.01	0.01		1.00

1 - 06/08/2023: Corn, silage

Field 1 Corn Silage

Sample and source description: Field 1 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,200.00	2,300.00	13,700.00		6.60
DL	0.01	0.01	0.01		1.00

10 - 10/05/2022: Oats, silage-soft dough

Field 10 Oats

Sample and source description: Field 10 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	3,100.00	22,400.00		10.90
DL	0.01	0.01	0.01		1.00

10 - 06/15/2023: Corn, silage

Field 10 Corn

Sample and source description: Field 10 Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	13,100.00	3,100.00	24,600.00		7.80
DL	0.01	0.01	0.01		1.00

2 - 12/01/2019: Alfalfa, hay

Alfalfa Mid Cut Field 2

Sample and source description: Alfalfa Mid Cut Field 2

Sample date: 08/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 72.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	29,500.00	3,100.00	13,800.00		8.30
DL	0.01	0.01	0.01		1.00

3 - 12/01/2019: Alfalfa, hay

Field 3 Alfalfa Green Chop

Sample and source description: Field 3 Alfalfa Green Chop

Sample date: 04/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 37.5 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	32,700.00	3,300.00	22,300.00		8.60
DL	0.01	0.01	0.01		1.00

Alfalfa Mid Cut Field 3

Sample and source description: Alfalfa Mid Cut Field 3

Sample date: 08/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,300.00	3,000.00	38,400.00		11.60
DL	0.01	0.01	0.01		1.00

4 - 10/29/2022: Oats, silage-soft dough

4 - 10/29/2022: Oats, silage-soft dough

Field 4 Oats

Sample and source description: Field 4 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	3,200.00	22,100.00		10.50
DL	0.01	0.01	0.01		1.00

4 - 06/08/2023: Corn, silage

Field 4 Corn Silage

Sample and source description: Field 4 Corn Silage

Sample date: 09/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 59.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,400.00	2,100.00	17,500.00		7.50
DL	0.01	0.01	0.01		1.00

5 - 10/08/2022: Oats, silage-soft dough

Field 5 Oats

Sample and source description: Field 5 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 68.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	19,500.00	3,200.00	22,100.00		10.50
DL	0.01	0.01	0.01		1.00

5 - 06/08/2023: Corn, silage

Field 5 Corn

Sample and source description: Field 5 Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 36.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,000.00	3,000.00	9,700.00		5.90
DL	0.01	0.01	0.01		1.00

6 - 12/01/2018: Alfalfa, hay

Field 6A Alfalfa Green Chop

Sample and source description: Field 6A Alfalfa Green Chop

Sample date: 04/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 45.9 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	31,900.00	2,500.00	21,000.00		9.30
DL	0.01	0.01	0.01		1.00

Field 6B Alfalfa Green Chop

Sample and source description: Field 6B Alfalfa Green Chop

Sample date: 04/13/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 33.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	27,200.00	2,500.00	20,500.00		8.10
DL	0.01	0.01	0.01		1.00

6 - 12/01/2018: Alfalfa, hay

Alfalfa Mid Cut Field 6

Sample and source description: Alfalfa Mid Cut Field 6

Sample date: 08/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 4.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	37,800.00	2,800.00	14,600.00		8.90
DL	0.01	0.01	0.01		1.00

7 - 12/01/2018: Alfalfa, hay

Alfalfa Mid Cut Field 7

Sample and source description: Alfalfa Mid Cut Field 7

Sample date: 08/01/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 8.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	35,600.00	3,300.00	24,300.00		10.20
DL	0.01	0.01	0.01		1.00

8 - 10/10/2022: Oats, silage-soft dough

Field 8 Oats

Sample and source description: Field 8 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 74.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,000.00	3,100.00	21,700.00		12.80
DL	0.01	0.01	0.01		1.00

8 - 06/10/2023: Corn, silage

Field 8 Corn

Sample and source description: Field 8 Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 51.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,800.00	1,800.00	8,600.00		7.70
DL	0.01	0.01	0.01		1.00

9 - 10/01/2022: Oats, silage-soft dough

Field 9 Oats

Sample and source description: Field 9 Oats

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 65.4 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	22,600.00	3,100.00	22,400.00		10.90
DL	0.01	0.01	0.01		1.00

9 - 06/12/2023: Corn, silage

Field 9 Corn

Sample and source description: Field 9 Corn

Sample date: 10/04/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.1 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	14,400.00	2,600.00	10,800.00		6.90
DL	0.01	0.01	0.01		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

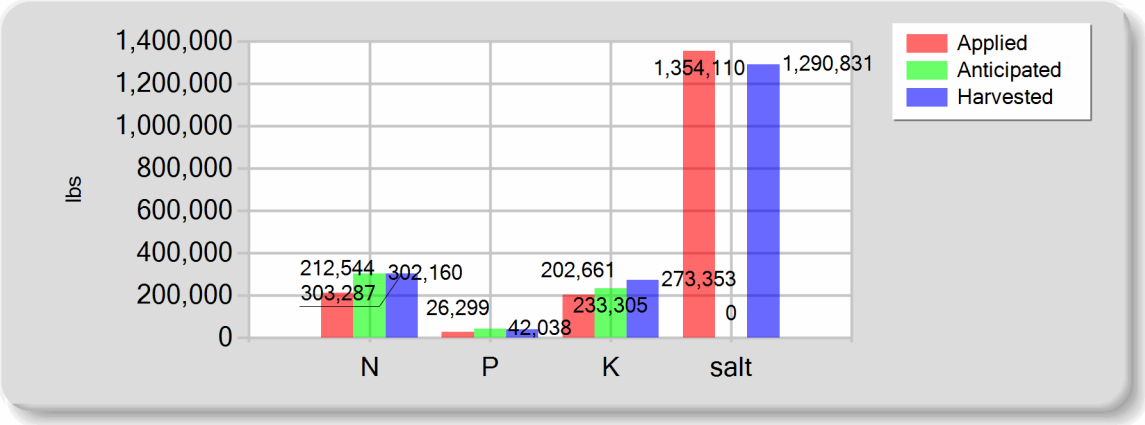
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

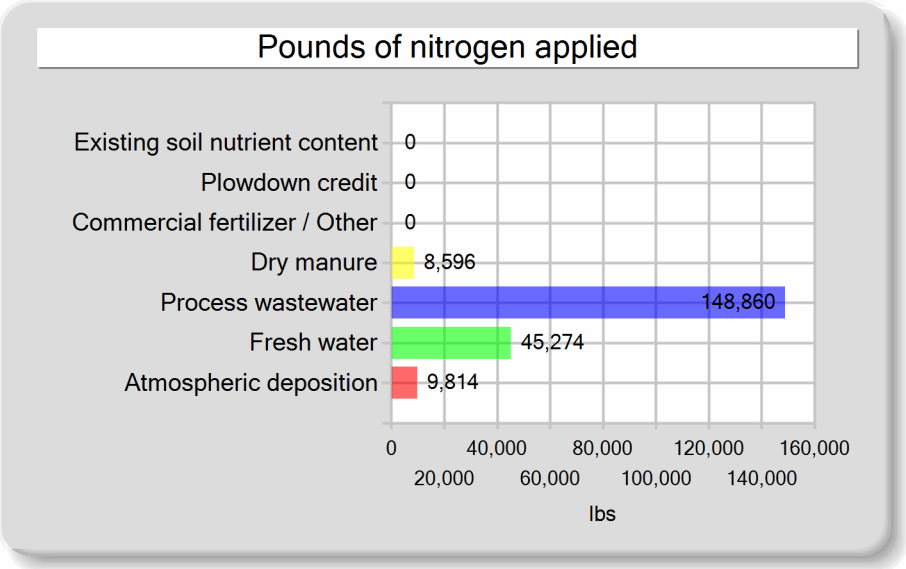
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

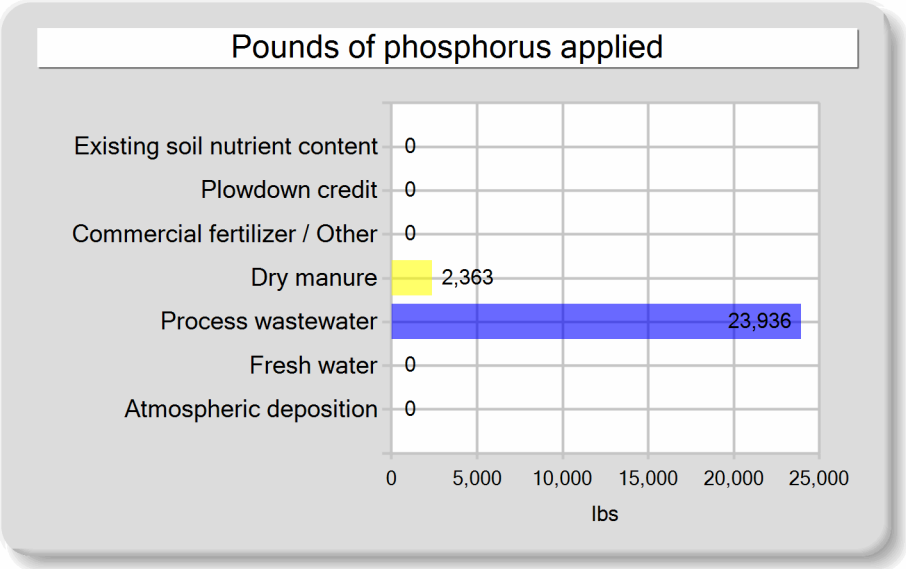
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	8,596.19	2,362.69	6,698.59	85,914.00
Process wastewater	148,860.00	23,935.90	195,962.46	1,168,582.73
Fresh water	45,273.85	0.00	0.00	99,613.60
Atmospheric deposition	9,814.00	0.00	0.00	0.00
Total nutrients applied	212,544.05	26,298.60	202,661.06	1,354,110.33
Anticipated crop nutrient removal	302,160.00	42,037.80	233,305.20	0.00
Actual crop nutrient removal	303,287.36	40,640.95	273,352.75	1,290,830.84
Nutrient balance	-90,743.31	-14,342.36	-70,691.69	63,279.49
Applied to removed ratio	0.70	0.65	0.74	1.05

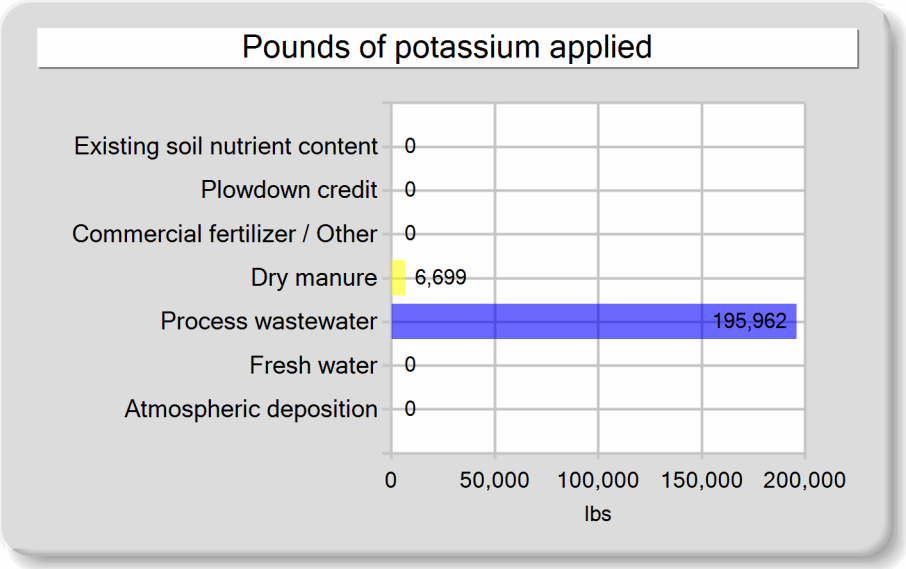
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

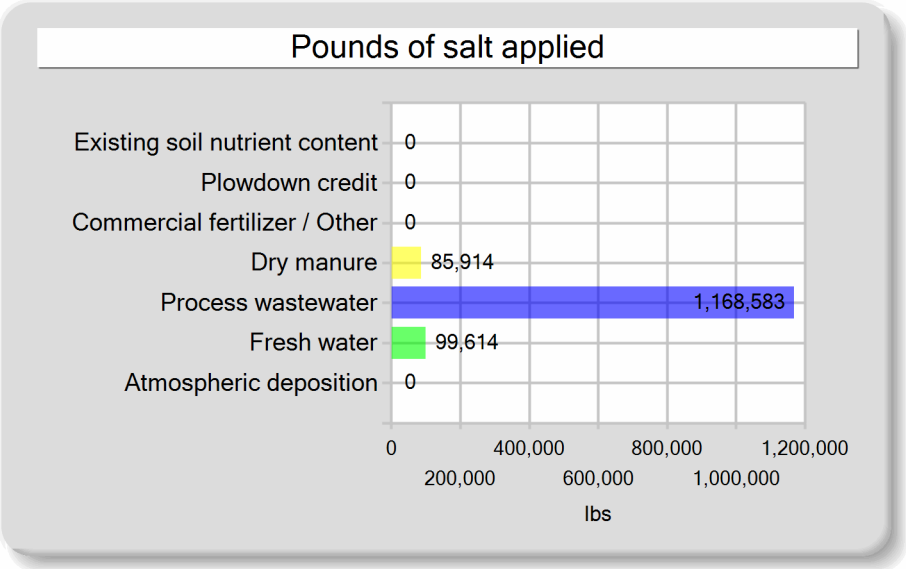


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

Report is submitted late due to hard financial times, lab results were not released by the lab before the July 1 deadline to complete the report

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

David Barroso	David Barroso
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Renee Barroso	David Barroso
PRINT OR TYPE NAME	PRINT OR TYPE NAME
Jul 15, 2024	Jul 15, 2024
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced CA 95341
00-0015955 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1009
Sampled By: M. Pedroso
Requested By: David Barroso
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Field NH_4-N mg/L	Total NH_4-N mg/L	pH at 25°C unit
1	DW - MB	12/13/23	11:45	474	0.47	5.1	ND		7.6
2	DW - House	12/13/23	11:48	703	0.70	13.1	ND		7.8
3	IW - F8 South	12/13/23	11:55	1420	1.42	6.3	ND		7.3
4	IW - F6	12/13/23	12:05	1240	1.24	9.6	ND		7.4
5	IW - F10	12/13/23	12:10	923	0.92	7.0	ND		7.6
6	IW - F4	12/13/23	12:20	1520	1.52	9.9	ND		7.5
7	IW - F7	12/13/23	12:30	553	0.55	5.5	ND		7.7
8	IW - F9	12/13/23	12:38	522	0.52	5.3	ND		7.5
9	IW - F8 North	12/13/23	12:45	581	0.58	5.7	ND		7.6

***** = Field NH_4-N not Taken.
ND = None Detected

Approved By: Scott M Friedland
Laboratory Director\Technical Manager
ELAP Certification #1595
A2LA Certification #6440.02

DELLAVALLE LABORATORY INC

Report of Dairy Well Water Analysis

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced CA 95341
00-0015955 08

E-mail:
Copy To: cas.labs@yahoo.com

Lab No.: 23L1009
Sampled By: M. Pedroso
Requested By: David Barroso
Submitted Date: 12/14/23
Reported Date: 12/21/23
Project:
Crop ID:

		Date Sampled	Time Sampled	EC µmhos/cm	EC mmhos/cm	NO_3-N mg/L	Nitrogen agricultural use Calcs calculated from nitrate-N lbs/AcFoot	lbs/AcInch
1	DW - MB	12/13/23	11:45	474	0.47	**5.1**	13.92	1.16
2	DW - House	12/13/23	11:48	703	0.70	**13.1**	35.76	2.98
3	IW - F8 South	12/13/23	11:55	**1420**	**1.42**	**6.3**	17.20	1.43
4	IW - F6	12/13/23	12:05	**1240**	**1.24**	**9.6**	26.21	2.18
5	IW - F10	12/13/23	12:10	**923**	**0.92**	**7.0**	19.11	1.59
6	IW - F4	12/13/23	12:20	**1520**	**1.52**	**9.9**	27.03	2.25
7	IW - F7	12/13/23	12:30	553	0.55	**5.5**	15.02	1.25
8	IW - F9	12/13/23	12:38	522	0.52	**5.3**	14.47	1.21
9	IW - F8 North	12/13/23	12:45	581	0.58	**5.7**	15.56	1.30

Water for Crop Use **Nutrient Management**	Total Salts	Total Salts	Nitrate Nitrogen
Low Levels	< 900	< 0.90	< 2.0
Significant Levels	900-2200	0.90-2.2	2.0-10.0
High Levels	2200+	2.2+	10.0+

Sampling abbreviations: H.B. = hose bib, S.P. = stand pipe, Dom. = domestic well, IR = irrigation well, AG = ag well.
Nitrate exceeds Water Quality MCL levels if results are above 10 mg/l nitrate-nitrogen NO3-N (equivalent to 45 mg/l nitrate, NO3).
Total Salt results are used to monitor changes of salt in the well aquifers between annual tests.
MCL = Maximum Contaminant Level according to the California Domestic Water Quality and Monitoring Regulations (Title 22)
MDL = Method Detection Limit; **RL** = Reporting Limit, mg/L = ppm.
SM = Standard Methods for the Examination of Water and Wastewater
EPA = Environmental Protection Agency methods used unless otherwise indicated.

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LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Samples in this Report

Lab ID	Sample	Matrix	Sampled By	Crop	Date Sampled
23L1009-01	DW - MB	Ag Water	M. Pedroso		12/13/2023 11:45
23L1009-02	DW - House	Ag Water	M. Pedroso		12/13/2023 11:48
23L1009-03	IW - F8 South	Ag Water	M. Pedroso		12/13/2023 11:55
23L1009-04	IW - F6	Ag Water	M. Pedroso		12/13/2023 12:05
23L1009-05	IW - F10	Ag Water	M. Pedroso		12/13/2023 12:10
23L1009-06	IW - F4	Ag Water	M. Pedroso		12/13/2023 12:20
23L1009-07	IW - F7	Ag Water	M. Pedroso		12/13/2023 12:30
23L1009-08	IW - F9	Ag Water	M. Pedroso		12/13/2023 12:38
23L1009-09	IW - F8 North	Ag Water	M. Pedroso		12/13/2023 12:45

Default Cooler Temperature on Receipt °C: 2.9
Containers Intact
COC/Labels Agree
Received On Ice

Notes and Definitions

Item	Definition
H	Hold Time Exceeded
MCL	Drinking Water Maximum Contaminant Level
ND	Analyte NOT DETECTED at or above the reporting limit.
NES	Not Enough Sample
*	Not Taken
RPD	Relative Percent Difference
%REC	Percent Recovery
Source	Sample that was matrix spiked or duplicated.

Scott McFriedland

Laboratory Director/Technical Manager

ELAP Certification #1595
A2LA Certification #6440.02

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1910 W. McKinley Ave Suite 110 Fresno, CA 93728 559-233-6129 www.dellavallelab.com

DELLAVALLE™
LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results

Sample: DW - MB
23L1009-01 (Water)

Sampled: 12/13/2023 11:45
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.47**	mmhos/cm	0.01	1		12/19/23 11:24	SM 2510 B		BEL0705
Electrical Conductivity umhos	**474**	umhos/cm	10.0	1		12/19/23 11:24	SM 2510 B		BEL0705
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 11:45	Field		BEL0652
Nitrate Nitrogen as NO3N	**5.1**	mg/L	0.1	1	10	12/18/23 17:04	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:24	SM 4500-H+	H	BEL0705
pH	**7.6**	units	1.0	1		12/19/23 11:24	SM 4500-H+	H	BEL0705

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LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results
(Continued)

Sample: DW - House
23L1009-02 (Water)

Sampled: 12/13/2023 11:48
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.70**	mmhos/cm	0.01	1		12/19/23 11:26	SM 2510 B		BEL0705
Electrical Conductivity umhos	**703**	umhos/cm	10.0	1		12/19/23 11:26	SM 2510 B		BEL0705
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 11:48	Field		BEL0652
Nitrate Nitrogen as NO3N	**13.1**	mg/L	0.1	1	10	12/18/23 17:24	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:26	SM 4500-H+	H	BEL0705
pH	**7.8**	units	1.0	1		12/19/23 11:26	SM 4500-H+	H	BEL0705

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LABORATORY INC

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6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results
(Continued)

Sample: IW - F8 South
23L1009-03 (Water)

Sampled: 12/13/2023 11:55
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.42**	mmhos/cm	0.01	1		12/19/23 11:27	SM 2510 B		BEL0705
Electrical Conductivity umhos	**1420**	umhos/cm	10.0	1		12/19/23 11:27	SM 2510 B		BEL0705
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 11:55	Field		BEL0652
Nitrate Nitrogen as NO3N	**6.3**	mg/L	0.1	1	10	12/18/23 17:44	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:27	SM 4500-H+	H	BEL0705
pH	**7.3**	units	1.0	1		12/19/23 11:27	SM 4500-H+	H	BEL0705

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DELLAVALLE™
LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results
(Continued)

Sample: IW - F6
23L1009-04 (Water)

Sampled: 12/13/2023 12:05
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.24**	mmhos/cm	0.01	1		12/19/23 11:29	SM 2510 B		BEL0705
Electrical Conductivity umhos	**1240**	umhos/cm	10.0	1		12/19/23 11:29	SM 2510 B		BEL0705
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:05	Field		BEL0652
Nitrate Nitrogen as NO3N	**9.6**	mg/L	0.1	1	10	12/18/23 18:03	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:29	SM 4500-H+	H	BEL0705
pH	**7.4**	units	1.0	1		12/19/23 11:29	SM 4500-H+	H	BEL0705

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LABORATORY INC

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6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results
(Continued)

Sample: IW - F10
23L1009-05 (Water)

Sampled: 12/13/2023 12:10
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.92**	mmhos/cm	0.01	1		12/19/23 11:30	SM 2510 B		BEL0705
Electrical Conductivity umhos	**923**	umhos/cm	10.0	1		12/19/23 11:30	SM 2510 B		BEL0705
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:10	Field		BEL0652
Nitrate Nitrogen as NO3N	**7.0**	mg/L	0.1	1	10	12/18/23 18:23	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:30	SM 4500-H+	H	BEL0705
pH	**7.6**	units	1.0	1		12/19/23 11:30	SM 4500-H+	H	BEL0705

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LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results
(Continued)

Sample: IW - F4
23L1009-06 (Water)

Sampled: 12/13/2023 12:20
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**1.52**	mmhos/cm	0.01	1		12/19/23 11:31	SM 2510 B		BEL0705
Electrical Conductivity umhos	**1520**	umhos/cm	10.0	1		12/19/23 11:31	SM 2510 B		BEL0705
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:20	Field		BEL0652
Nitrate Nitrogen as NO3N	**9.9**	mg/L	0.1	1	10	12/18/23 18:43	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:31	SM 4500-H+	H	BEL0705
pH	**7.5**	units	1.0	1		12/19/23 11:31	SM 4500-H+	H	BEL0705

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LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results
(Continued)

Sample: IW - F7
23L1009-07 (Water)

Sampled: 12/13/2023 12:30
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.55**	mmhos/cm	0.01	1		12/19/23 11:33	SM 2510 B		BEL0705
Electrical Conductivity umhos	**553**	umhos/cm	10.0	1		12/19/23 11:33	SM 2510 B		BEL0705
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:30	Field		BEL0652
Nitrate Nitrogen as NO3N	**5.5**	mg/L	0.1	1	10	12/18/23 19:03	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:33	SM 4500-H+	H	BEL0705
pH	**7.7**	units	1.0	1		12/19/23 11:33	SM 4500-H+	H	BEL0705

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LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results
(Continued)

Sample: IW - F9
23L1009-08 (Water)

Sampled: 12/13/2023 12:38
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.52**	mmhos/cm	0.01	1		12/19/23 11:48	SM 2510 B		BEL0706
Electrical Conductivity umhos	**522**	umhos/cm	10.0	1		12/19/23 11:48	SM 2510 B		BEL0706
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:38	Field		BEL0652
Nitrate Nitrogen as NO3N	**5.3**	mg/L	0.1	1	10	12/18/23 22:42	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:48	SM 4500-H+	H	BEL0706
pH	**7.5**	units	1.0	1		12/19/23 11:48	SM 4500-H+	H	BEL0706

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LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Sample Results
(Continued)

Sample: IW - F8 North
23L1009-09 (Water)

Sampled: 12/13/2023 12:45
Sampled By: M. Pedroso

Analyte	Result	Units	Reporting Limit	DIL	DW MCL	Date/Time Analyzed	Method	Notes	Batch
Electrical Conductivity	**0.58**	mmhos/cm	0.01	1		12/19/23 11:50	SM 2510 B		BEL0706
Electrical Conductivity umhos	**581**	umhos/cm	10.0	1		12/19/23 11:50	SM 2510 B		BEL0706
Ammonia (as N)	ND	mg/L	0.00	1		12/13/23 12:45	Field		BEL0652
Nitrate Nitrogen as NO3N	**5.7**	mg/L	0.1	1	10	12/18/23 19:23	EPA 300.0		BEL0686
Temperature	**25.0**	units	0.0	1		12/19/23 11:50	SM 4500-H+	H	BEL0706
pH	**7.6**	units	1.0	1		12/19/23 11:50	SM 4500-H+	H	BEL0706

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DELLAVALLE™
LABORATORY INC

Joe & Renee Barroso Dairy
6902 E. Le Grand Rd
Merced, CA 95341

Account# 00-0015955
Account Manager: Ben Nydam
Submitted By: David Barroso

Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Quality Control

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0686										
Blank (BEL0686-BLK1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
Blank (BEL0686-BLK3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	ND		0.1	mg/L						
LCS (BEL0686-BS1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		101	90-110		
LCS (BEL0686-BS2)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.1		0.1	mg/L	5.000		102	90-110		
Duplicate (BEL0686-DUP1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	5.2		0.1	mg/L		5.1			0.974	10
Duplicate (BEL0686-DUP2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	5.3		0.1	mg/L		5.3			0.530	10
Matrix Spike (BEL0686-MS1)		**Source: 23L1009-01**			Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.1	106	90-110		
Matrix Spike (BEL0686-MS2)		**Source: 23L1009-08**			Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.4		0.1	mg/L	5.000	5.3	102	90-110		
Reference (BEL0686-SRM1)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.2			mg/L	10.00		102	90-110		
Reference (BEL0686-SRM2)					Prepared & Analyzed: 12/18/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		
Reference (BEL0686-SRM3)					Prepared: 12/18/2023 Analyzed: 12/19/2023					
Nitrate Nitrogen as NO3N	10.3			mg/L	10.00		103	90-110		

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Received: 12/14/2023 14:20
Reported: 12/21/2023 12:16

Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0705										
Blank (BEL0705-BLK1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	5.6		1.0	units						
Blank (BEL0705-BLK2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.6		1.0	units						
Blank (BEL0705-BLK3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	7.7		1.0	units						
Duplicate (BEL0705-DUP1)		**Source: 23L1008-04**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	2.75		0.01	mmhos/cm		2.72			1.10	10
pH	7.6		1.0	units		7.6			0.264	10
Electrical Conductivity umhos	2750		10.0	umhos/cm		2720			1.10	10
Duplicate (BEL0705-DUP2)		**Source: 23L1009-07**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	0.55		0.01	mmhos/cm		0.55			0.343	10
pH	7.7		1.0	units		7.7			0.259	10
Electrical Conductivity umhos	554		10.0	umhos/cm		553			0.343	10
Reference (BEL0705-SRM1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	430			umhos/cm	426.0		101	90-110		
Reference (BEL0705-SRM2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0705-SRM3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1020			umhos/cm	1000		102	90-110		
Electrical Conductivity umhos	1020			umhos/cm	1000		102	90-110		
Reference (BEL0705-SRM4)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEL0705-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1050			umhos/cm	1000		105	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0705 (Continued)										
Reference (BEL0705-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity umhos	1050			umhos/cm	1000		105	90-110		
Reference (BEL0705-SRM6)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0705-SRM7)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0705-SRM8)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706										
Blank (BEL0706-BLK1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	ND		0.01	mmhos/cm						
Temperature	25.0		0.0	units						
pH	5.6		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0706-BLK2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
pH	7.9		1.0	units						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
Blank (BEL0706-BLK3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Temperature	25.0		0.0	units						
Electrical Conductivity	ND		0.01	mmhos/cm						
Electrical Conductivity umhos	ND		10.0	umhos/cm						
pH	6.8		1.0	units						
Duplicate (BEL0706-DUP1)		**Source: 23L1017-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1.74		0.01	mmhos/cm		1.73			0.403	10
pH	7.7		1.0	units		7.8			0.258	10
Electrical Conductivity umhos	1740		10.0	umhos/cm		1730			0.403	10
Duplicate (BEL0706-DUP2)		**Source: 23L1088-01**			Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	0.52		0.01	mmhos/cm		0.52			0.116	10
pH	6.7		1.0	units		6.8			0.297	10
Electrical Conductivity umhos	517		10.0	umhos/cm		517			0.116	10
Reference (BEL0706-SRM1)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	440			umhos/cm	426.0		103	90-110		
Reference (BEL0706-SRM2)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	7.5			units	7.520		100	67021-101.32		
Reference (BEL0706-SRM3)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1040			umhos/cm	1000		104	90-110		
Electrical Conductivity umhos	1040			umhos/cm	1000		104	90-110		
Reference (BEL0706-SRM4)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity	1030			umhos/cm	1000		103	90-110		

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Quality Control
(Continued)

Analyte	Result	Qual	Reporting Limit	Units	Spike Level	Source Result	%REC	%REC Limits	RPD	RPD Limit
Batch: BEL0706 (Continued)										
Reference (BEL0706-SRM5)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
Electrical Conductivity umhos	1030			umhos/cm	1000		103	90-110		
Reference (BEL0706-SRM6)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM7)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		101	97.5-102.5		
Reference (BEL0706-SRM8)					Prepared: 12/18/2023	Analyzed: 12/19/2023				
pH	4.0			units	4.000		100	97.5-102.5		

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12/14/23 14:20 23L1009

WATER WORK REQUEST

Bill To: Acct No. 15955 Cons. 8

Purchase Order No. Results Needed By

Client: Joe & Renee ~~J & R~~ Barroso ~~& Sons~~ Dairy ~~LP~~

Address: 6902 E Le Grand Road

City, State, Zip: Merced, CA 95340

Phone: Fax:

Cell/Email:

Copy to: Cardoso Ag Sevices - cas.labs@yahoo.com

Requested by: David Barroso - (209) 652-6437

Ranch:

Date sampled: 12/13/23

Sampled by: M Pedroso

[X] QA/QC Document [X] Copy of Chain [ ] RWQCB

DELLAVALLE LABORATORY, INC.

1910 W. McKinley Avenue, Suite 110 • Fresno, CA 93728

www.dellavallelab.com 559 233-6126 • 800 228-9896 • Fax 559 268-8174

No. of Samples 9 No. Bottles 9

Water Type: [ ] Drinking [ ] Wastewater

[X] Ag Water [ ] Ground Water [ ] Mon. Well

[ ] Supply Water [ ] Other

Analysis and Bottles Required: *(Please Indicate Analysis)*

(X) DWW1: (EC, pH, NO_3-N, **NH_4-N Field Test)**
(1) l L plastic, unpreserved (white)

( ) DWW2: (DWW1 Plus SO_4, CO_3, HCO_3, Cl, Ca, Mg, Na, TDS)
(1) l L plastic, unpreserved (white)

( ) DCW1: (EC, NO_3-N, TDS)
(1) l L plastic, unpreserved (white)

( ) DPW1: (EC, pH, NO_3-N, NH_4-N, TKN, TDS, TP, TK)
(1) l L plastic, unpreserved (white)

( ) DPW2: (DPW1 Plus Ca, Mg, Na, HCO_3, CO_3, SO_4, Cl)
(1) l L plastic, unpreserved (white)

( ) Other

DESCRIPTION OF SAMPLES		Date Sampled	Time Sampled	Field NH4-N (mg/L)	Received Temp °C
1. DW-MB	Sampled From:	12/13/23	1145A	0	2.9
2. DW-House	Sampled From:	12/13/23	1148A	0	3.4
3. IW-F8 South	Sampled From:	12/13/23	1155A	0	4.9
4. IW-F6	Sampled From:	12/13/23	1205P	0	5.5
5. IW-F10	Sampled From:	12/13/23	1210P	0	4.2
6. IW-F4	Sampled From:	12/13/23	1220P	0	3.6
7. IW-F7	Sampled From:	12/13/23	1230P	0	4.1
8. IW-F9	Sampled From:	12/13/23	1238P	0	3.3
9. IW-F8 North	Sampled From:	12/13/23	1245P	0	3.4
10.	Sampled From:				

CHAIN OF CUSTODY

Carrier	Signature	Company	Received (Date/Time)	Relinquished (Date/Time)
First	MPed	CAS Inc	12/13/23 1148A	12/14/23 12p
Second	CSM	DLI	12/14/23 12pm	12/14/23 2:15pm
Third				
Fourth	[illegible]	DLI	12/14 14:20	

I guarantee that as the client, or on behalf of the client named I have the authority to contract the above requested services. Should it be found that I do not have such authority, I agree to be personally liable for all costs and, if there should be action against me for this breach, reasonable attorneys' fees. It is understood that payment is expected to be cash with samples unless terms have been previously arranged. Terms are net 30 days; overdue accounts will be charged a dated damage fee of 2% per month (annually 24 %) or $5.00 per month whichever is greater.

If payment is not made when due and a legitimate dispute exists concerning the product or services of Dellavalle Laboratory, Inc., it will be submitted to mediation under the Rules and Procedures of Creative Alternative to Litigation, Inc. (cal). If the dispute is not resolved in mediation, then the dispute will be submitted to binding arbitration through cal under its Rules and Procedures. The prties will equally bear the costs of mediation/arbitration. If, however, the mediator declares that no legitimate dispute exists, then debtor will pay all mediation and arbitration costs, and in the event of arbitration, reasonable attorneys' fees of Dellavalle Laboratory.

Invoicing Information:

Sampling Hrs ___ Miles ___ Consulting ___

Amt Paid ___ Rec By ___ Check No. ___

Shipping

$ ___ In

$ ___ Out

Date ___

Signature

Sample received in cooler with ice?

[ ] Yes [ ] No

IR Thermometer SN: 221314357
Correction Factor: 0°C
Calibration Due
Location: Labor

Shipping Information: Shipped In ☐ Picked-Up ☒ Walk In ☐ DLI Sampler ☐ Other ☐ _____

Container: Ice Chest ☒ Box ☐ None ☐ **Refrigerant:** Wet Ice ☒ Blue Ice ☐ None ☐

Samples Preserved with HNO_3 or H_2SO_4 were: ☐ Received Preserved ☐ Preserved Upon Receipt at Laboratory

	Type of Container(s) Received	Sample Number 1	2	3	4	5	6	7	8	9	10
	Sample Containers for Internal (DLI) Use *(Containers that go into the Lab)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	* pH Value										
	250 mL H_2SO_4 (Yellow) Plastic										
	* pH Value										
	500 mL unpreserved (White) Plastic										
	1 L unpreserved (White) Plastic	1	1	1	1	1	1	1	1	1	
	1 L unpreserved (BOD) (Purple) Plastic										
"	500mL unpreserved (White) Glass										
	PO4-P Kit										
	Other:										
	Sample Containers for Subcontracted ("Send Out") Analyses *(Containers that go in the Subcontract ("Send Out") Refrigerator)*										
Plastics	100 mL sterile plastic $Na_2S_2O_3$ (Green)										
	250 mL unpreserved (White) Plastic										
	250 mL HNO_3 (Red) Plastic										
	250 mL H_2SO_4 (Yellow) Plastic										
	500 mL HNO_3 (Red)										
	1 L unpreserved (White) Plastic										
	1 L unpreserved (BOD) (Purple) Plastic										
	1 L HNO_3 (Red)										
VOA Vials	40 mL AG VOA, $Na_2S_2O_3$ + MCAA										
	40 mL AG VOA, $Na_2S_2O_3$										
	40mL AG VOA unpreserved (White) (Set of 3										
	40 mL AG VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
	40mL VOA, H_3PO_4 (Set of 3)										
	40 mL VOA, HCl (Blue) (Set of 3)										
	40 mL VOA, $Na_2S_2O_3$ (Green) (Set of 3)										
Glass	250 mL AG unpreserved (White)										
	250 mL AG H_2SO_4 (Yellow)										
	250 mL AG $Na_2S_2O_3$ (Green)										
	250 mL AG $Na_2S_2O_3$ + MCAA										
	500 mL glass unpreserved (White)										
	500 mL AG HCl (Blue)										
	1 L AG unpreserved (White)										
	1 L AG H_2SO_4 (Yellow)										
	1 L AG $Na_2S_2O_3$ (Green)										
	1 L AG HCl (Blue)										
Special	Cr^{6+} - 50mL Plastic w/Borate/HCO_3/CO_3										
	Cyanide - 500 mL NaOH										
	Asbestos - 1L P wrapped in foil (Set of 2)										
	Sulfide - 1 L **AG** or **P** NaOH + ZnAc										
	Chlorite/Bromate - 250 mL AG with EDA										
	HAA5 - 250mL AG Ammonium Chlorite										
	DO KIT										
	Other:										
	Other:										

12/14/23 14:20 23L1009

SEND OUT